Trend research — 2026-04-19 · incr 27

Structure is collinear with strength — don't bolt it onto the live gate

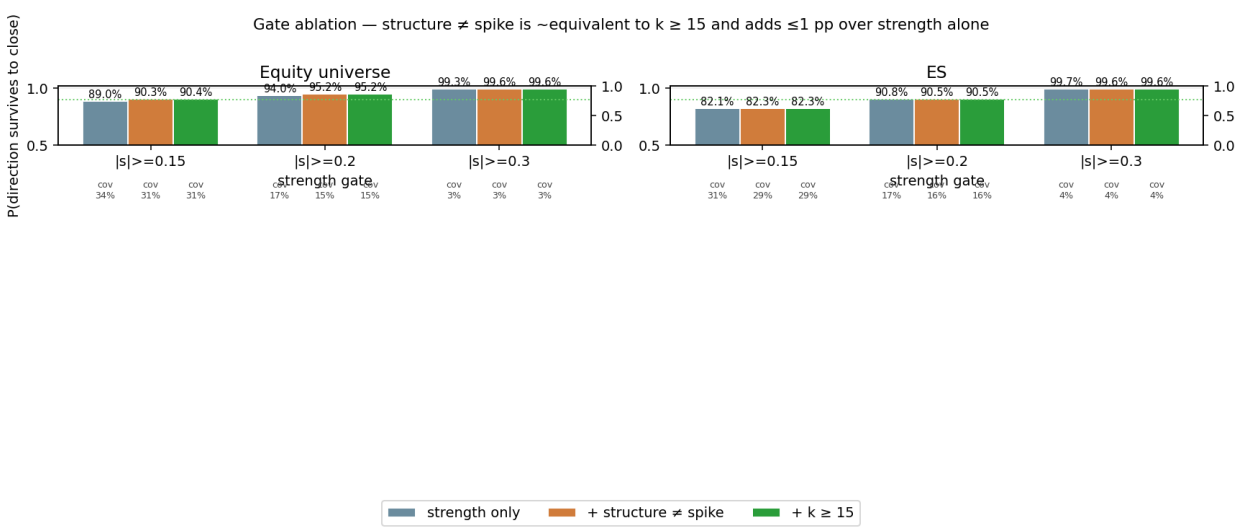
TL;DR

The **+15 pp** channel-vs-spike survival gap on equities (and **+9 pp** on ES) is a composition artifact. Once we condition on **|strength| ≥ 0.15**, adding a **structure ≠ spike** filter buys only **+1.3 pp** survival on equities and **+0.2 pp** on ES. The **structure ≠ spike** and **k ≥ 15** filters are **statistically interchangeable** — 100 % of spike-labelled directional observations on both populations happen at bars 10-14. Negative result: the incr-23 rule *"|strength| ≥ 0.15 AND bar ≥ 20"* is already near-optimal. Don't bolt structure onto the live gate.

Headline numbers

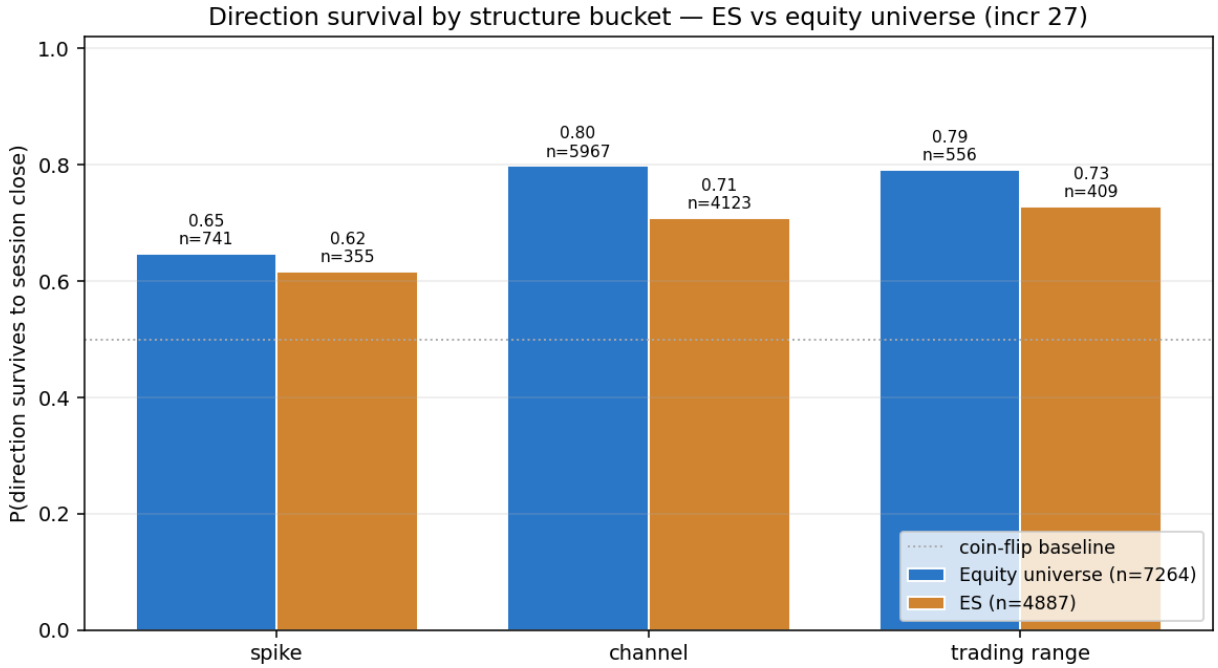
- Source: 200 equity RTH sessions (incr 23 trajectory, 7,264 directional obs) + 101 ES.c.0 RTH sessions (incr 26 trajectory, 4,887 directional obs). No new data fetch.
- **Bucket effect (raw):** equity channel **0.798** vs spike **0.648** (**+15 pp**). ES channel **0.708** vs spike **0.617** (**+9 pp**). Looks decisive until you ablate.
- **Ablation on equity at |s|≥0.15:** strength-only 89.0 % → + structure ≠ spike 90.3 % (**+1.3 pp**) → + k ≥ 15 90.4 %. Structure is within **0.2 pp** of the bar-count rule.
- **Ablation on ES at |s|≥0.20:** strength-only 90.8 % → + structure ≠ spike 90.5 % (**-0.3 pp**). On ES the structure filter is actively *unhelpful* once strength is conditioned.
- **100 % of spike directional obs land in bars 10-14** on both populations. **structure ≠ spike** and **k ≥ 15** are the same filter with a different name.
- **channel and trading_range are near-identical** (79.8 % vs 79.1 % equity; 70.8 % vs 72.9 % ES). Non-spike is the only meaningful structural boundary.

Headline figure — gate ablation



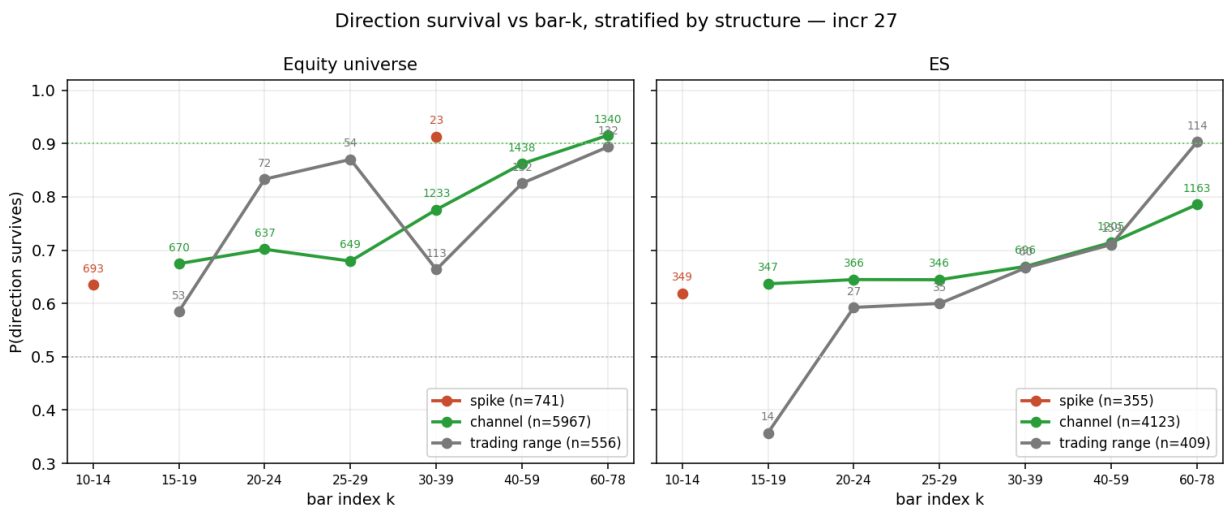
Per asset: three strength cuts × three gate variants. 'strength only', '+ structure ≠ spike', and '+ k ≥ 15' come out within 1-2 pp of each other at every cut. Structure is collinear with strength once we condition.

Bucket bars (raw) — what tempted the hypothesis



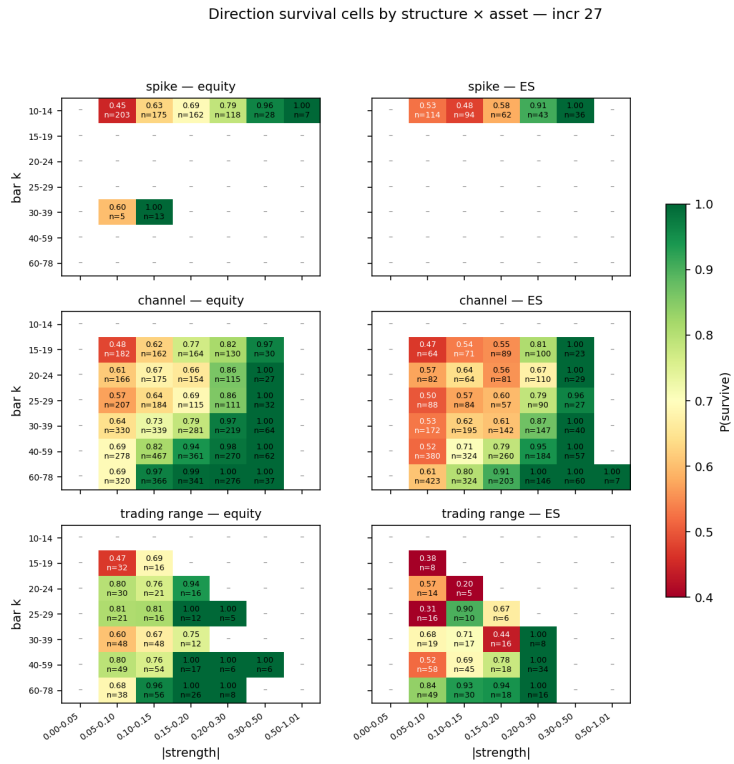
channel >> spike by 15 pp on equity, 9 pp on ES. Raw cut, not strength-conditioned — this is the naive reading. The ablation figure above tells the real story.

Per-k survival trajectories by bucket



channel rises monotonically in both populations. ES channel asymptote is 13 pp below equity channel — a confirmation of incr 26's 'ES is noisier' finding at the structure-conditioned level. Spike curves terminate early because spike is only emitted at bar 10-14.

Cell heatmap grid (3 × 2)



channel and trading_range panels show the familiar bar-k × |strength| separation. Spike panels have only the top row populated (bars 10-14) — confirming visually that structure is not a free k-independent feature.

Recommendation

- **Keep the incr-23 proposed rule:** *emit high-confidence direction iff |strength| ≥ 0.15 AND k ≥ 20*. Already near-optimal.
- **Do NOT add structure ≠ spike as a gate input.** It's collinear with k ≥ 15 (5 bars earlier) and adds ≤ 1.3 pp survival at the cost of extra plumbing.
- **Asset-class strength knob:** equity needs |s| ≥ 0.15 for 90 %; ES needs |s| ≥ 0.20. This is the only knob that matters.
- **Emit structure as a display-only label** on the dashboard card. Spike→channel transition is human-interpretable context; don't make it a gate.

Mistakes avoided this pass

- **Didn't ship the bucket-chart story.** A +15 pp gap looks decisive until you condition on the other variables. Always ablate before shipping a feature.
- **Didn't treat 'interpretable label' as free signal.** Structure reads well on a dashboard, but adding it to the gate buys no survival and costs cognitive load.
- **Didn't evaluate on survival alone.** At |s|≥0.30 every variant converges to 99 %+. The discriminating band is |s| ∈ [0.15, 0.20] where coverage is still useful.
- **Didn't conflate 'spike is bad' with 'spike hurts direction'.** Spike observations have low survival because they fire at bar 10-14 before confidence accrues — not because spike-ness is intrinsically toxic.

- **Counted the zero cells.** structure = spike has zero obs at bar ≥ 15 on ES. Any gate 'conditioning on structure' at late bars is a no-op.

What's next — requires Will's nod

- **Wire the simpler gate:** one strength cut keyed on asset class (0.15 equity, 0.20 ES) + k ≥ 20. No structure input.
- **Still pending from prior incrs:** ALWAYS_IN_WINDOW 5→10 (incr 20), MAJORITY_TREND_BAR_FLOOR 0.40→0.25 (incr 18), OPEN_EXTREME_THRESHOLD 0.15→0.25 (incr 21), htf_alignment silent-zero docstring (incr 19).
- **Pattern Lab WR overlay** — still blocked on Pattern Lab DB backfill.

Findings site: https://github.com/zerosumsystems-ui/iPhone-/tree/main/trend-classification